

STAT0006 ICA 3

Group 39

Group project (team of 4)

Part 1: Normal linear model

Analysis of property prices

Introduction to the data

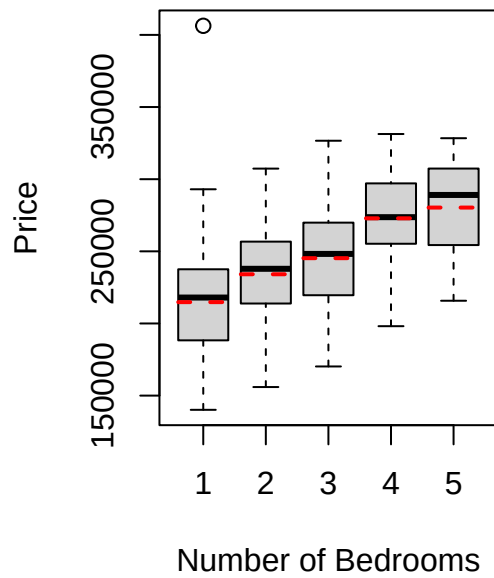
An experiment was conducted to explore how the price of properties in Statsville might be affected by different factors.

The dataset contains entries for 500 properties with no missing values. There is a total of 11 variables, including the price of the property. The mean price is SD-232590.4 and the standard deviation is SD-37803.63. The prices range from SD-140155 to SD-406305.

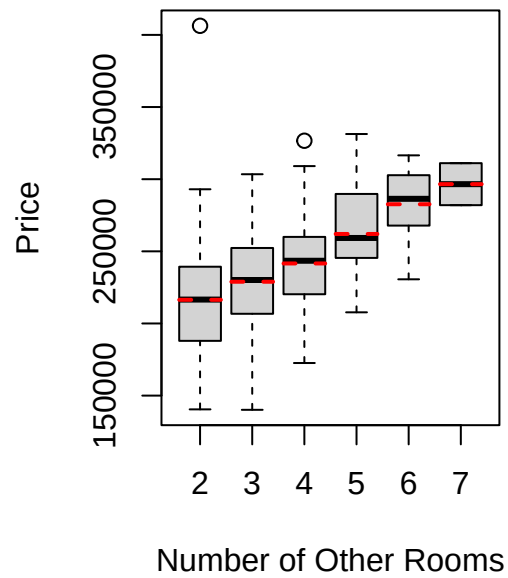
We will plot every variable against **price**. Some numerical variables only take a short range of discrete values and will be treated as categorical. In the presented boxplots, the red line shows the mean.

We start by looking at how different measures of size affect prices. Looking at the boxplots it seems that properties with more rooms, tend to have higher prices. The scatterplot shows a slight positive relationship between area and price. There is one property that stands out in all plots as it has a low number of rooms and bedrooms but has a disproportionately large price (SD-406305). However, looking at the scatterplot, it is clear that this property has a much higher area ($300m^2$), which explains the higher price. The data shows that bigger size, whether its measure by rooms or area, tends to increase the price of the property.

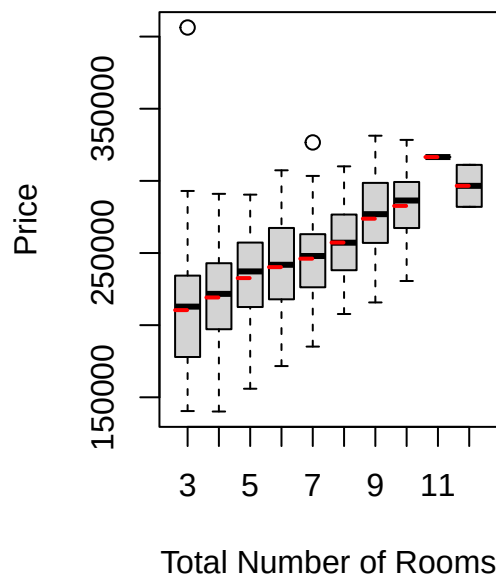
Price by Number of Bedroom:



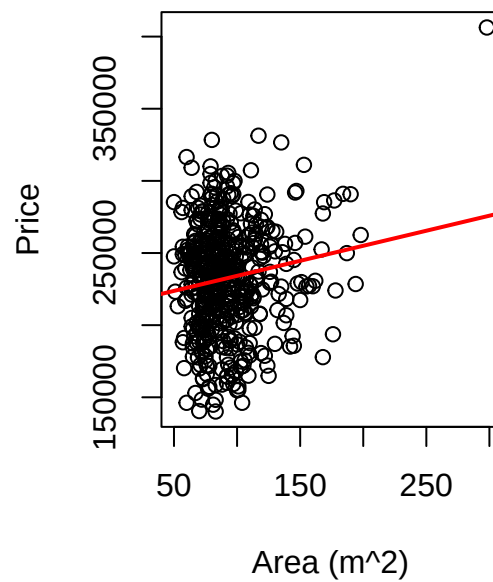
Price by Number Other Room



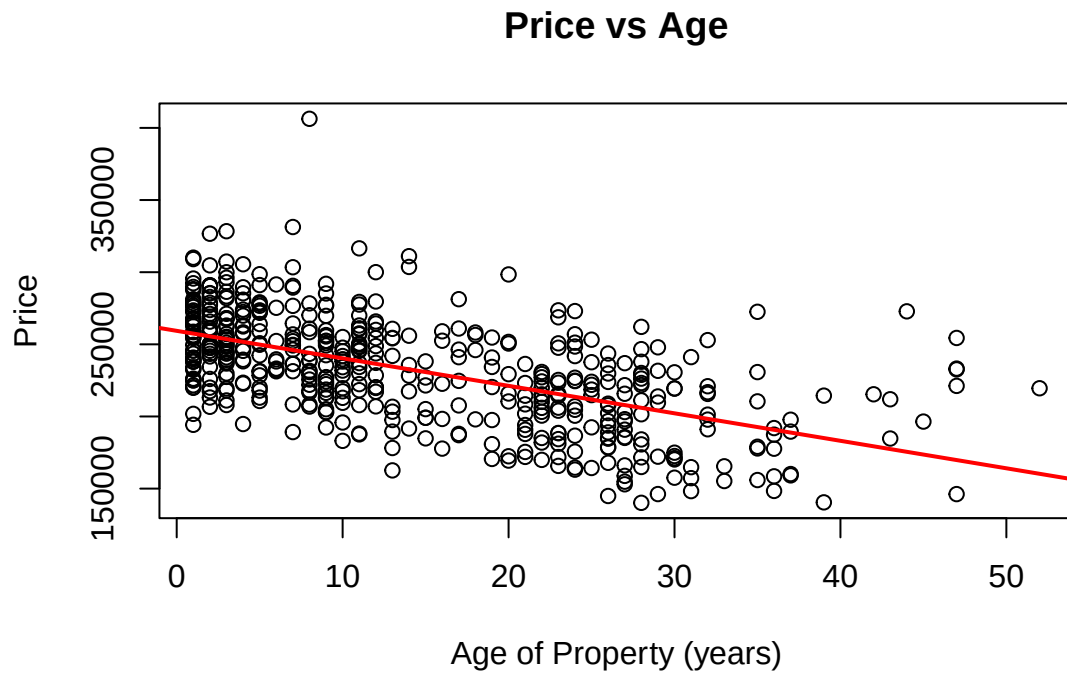
Price by Total Number of Room



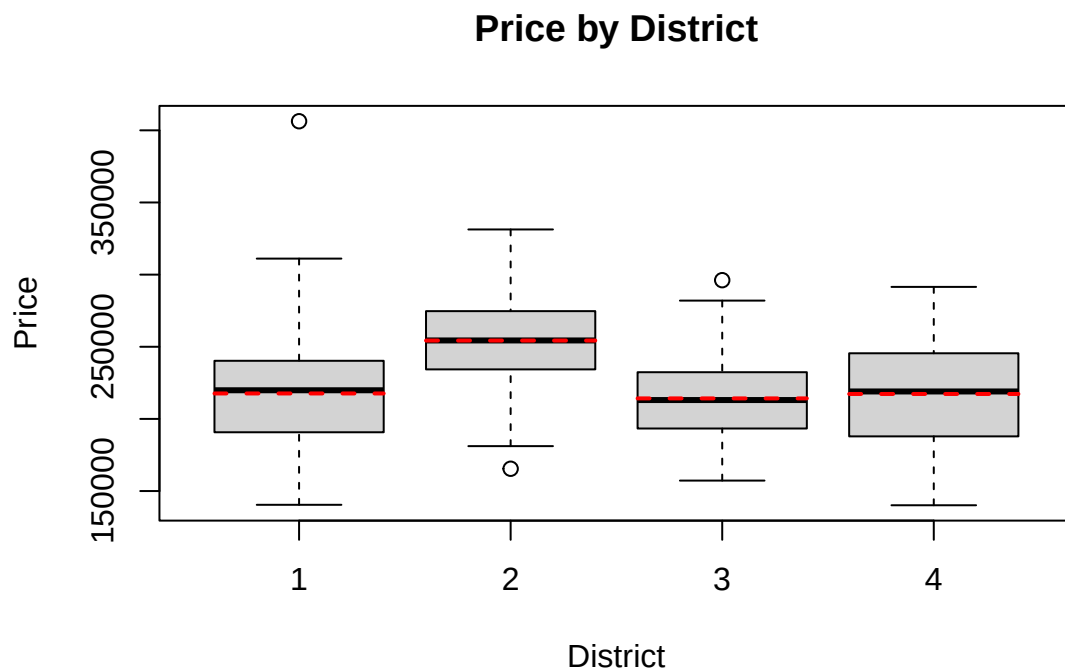
Price vs Area



Below there is a plot showing the relationship between the age of the property in years and price. The mean age for a property is 14.044 years. The plot indicates that price tends to decrease as the property becomes older.

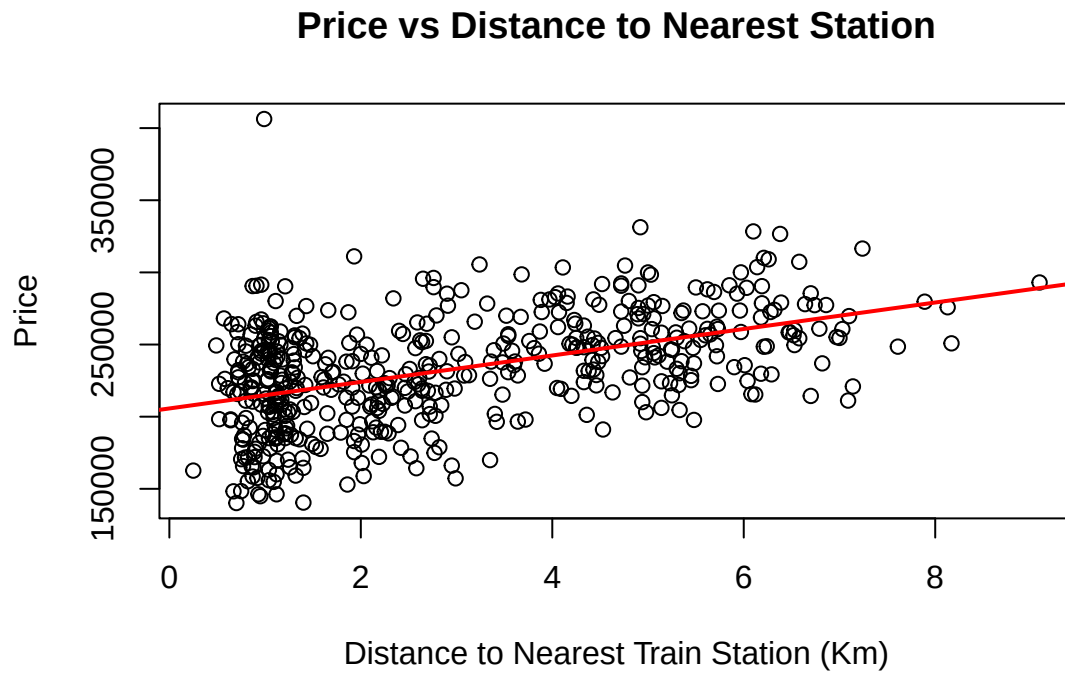


Below there's a boxplot showing the relationship between price and the district where the property is located (1-4). The mean prices for each district form 1-4 are 217627.6, 254298.2, 214223.9 and 217277.6. The plot shows that properties in district 2 have the highest average prices, compared to the others, which are similar.

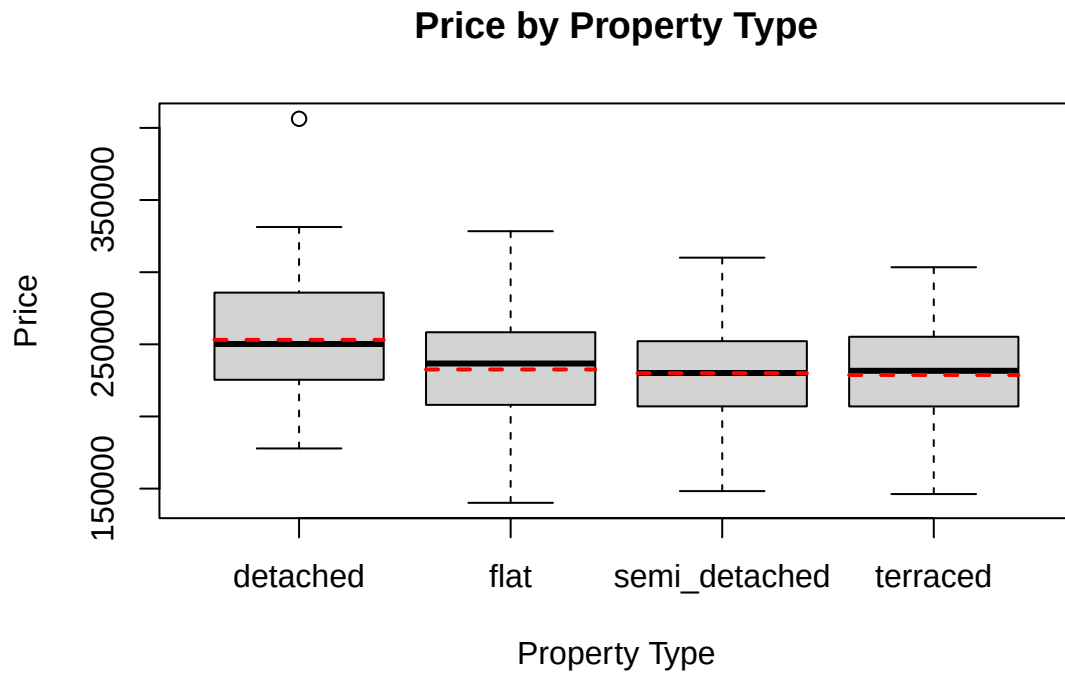


Furthermore, the plot below shows the relationship between price and the distance to the nearest train

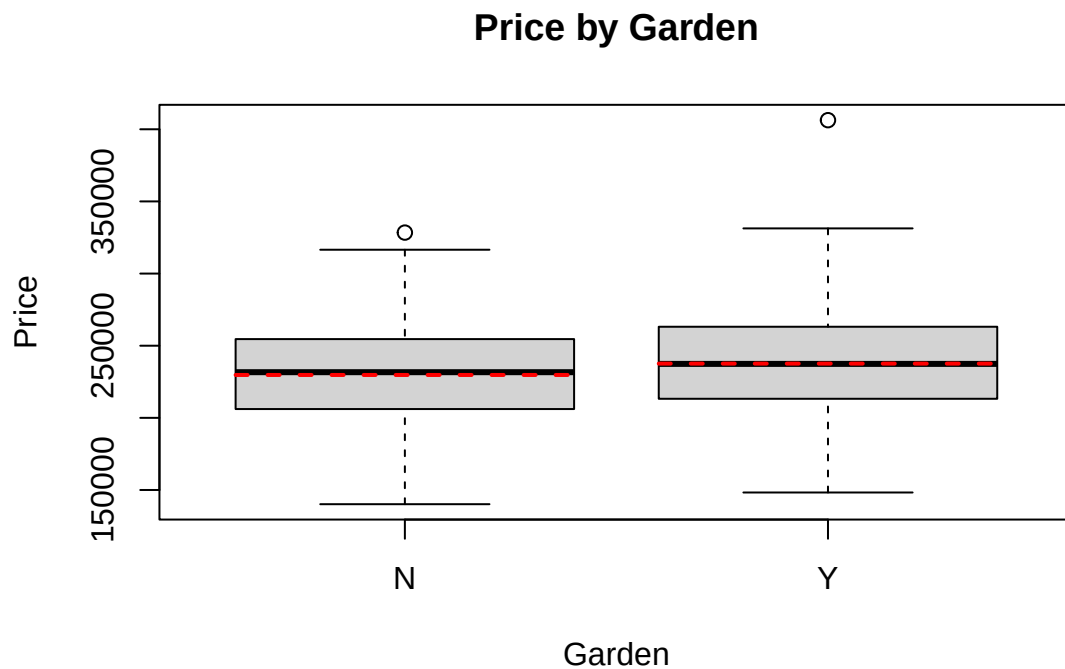
station in km. The mean distance to the nearest station is 2.93 km. The plot shows a positive relationship, meaning that price tends to be higher for properties further away from train stations



The types of properties and their average prices are: terraced (SD-228752.2), flat (SD-232577.1), detached (SD-253086.4) or semi-detached (SD-230022.2). The boxplot shows that the price for detached homes tends to be the highest, followed by flats, that have slightly higher average prices than semi-detached and terraced, but have higher range/variability. Semi-detached and terraced have similar average prices and variability.

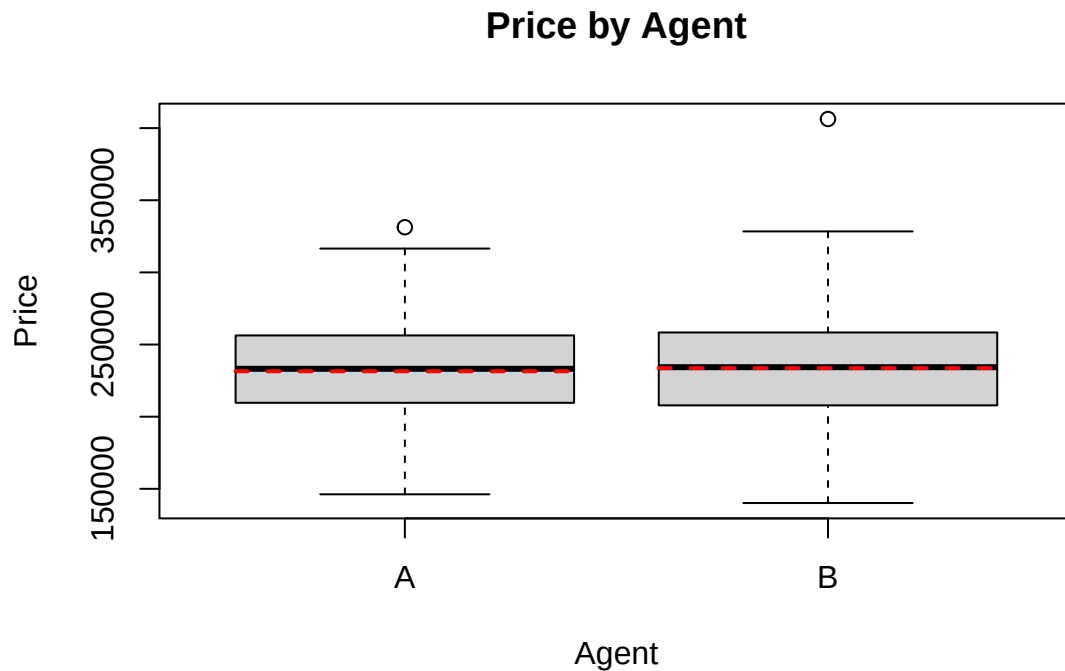


Below there is a boxplot showing the relationship between price and whether the property has a garden. The mean price for properties with a garden is 237588.6 and 229729.8 for properties without one. The plot shows higher prices for properties with gardens. However, the difference doesn't seem to be significant.



Finally, we can compare the price with the agent who sold the property, as shown in the boxplot below.

Agent A has a mean price of 231470.9 and B has mean 233666.0. The plot shows no significant difference between the agents.



Overall, most variables have shown to be significant when analyzing prices, with the exception of the variable **agent**. Nevertheless, this might still be significant, so all variables should be considered when building a model.

Model building

We decided that we should avoid using an algorithmic approach and simply use logic and our understanding of the context of the data in order to create an initial model and then simply manipulate it, making tactical adjustments based on our extensive EDA in order to find the best fit. By incorporating the most relevant patterns, we construct a model that captures the underlying structure of the data more effectively, improving its interpretability.

Setup of the initial model We saw a clear relationship between the selling price (the response variable) and all of the different covariates, except between **price** and **garden**, and **price** and **agent**, which impact doesn't seem significant. We will then proceed to build our initial model (`model_setup`) with all of the covariates mentioned to have a significant relationship with our response variable. We now have a suggested smaller set of covariates to work with, and to start exploring potential models.

```
model_setup <- lm(price ~ bedroom_count + other_count + total_count + age + area  
                  + station + district + type, data = prices)
```

Observations and corrections Firstly, as district values represent distinct geographic regions we will use the `as.factor()` function to change it to categorical. Treating it as numerical would incorrectly imply a linear relationship between districts, making misleading results.

Secondly, we detect collinearity between `total_count` and other covariates, which we can immediately deduce that are `bedroom_count` and `other_count`, as total count is the sum of the other two covariates. We will then proceed to compare two models, one with the two other covariates (`model_1`) and the other with the covariate total count (`model_2`).

After comparing the two models, by following the parsimony principle, we decided to prioritize simplicity and efficiency. Therefore we selected as preferred model, `model_2`, with the covariate `total_count` as the predictor. While `model_1` achieves a slightly higher adjusted R squared (0.8951 vs 0.8839), the difference is relatively small. We chose the option with the reduced number of covariates, as it adequately explains the variation in property value while avoiding the extra parameter.

We also realise that in both summaries, the estimated intercept is around SD 160,000, but we don't need to be concerned about it. The intercept value corresponds to the expected selling price when all categorical covariates are set to 0, this isn't a realistic property scenario.

By trying to further simplify the model, we use logic to propose the possible correlation between the covariates `total_count` and `area`. Having more rooms could imply having more square meters in the house. However, the correlation coefficient between the two covariate is -0.087. As this result is close to 0, we conclude that there is no correlation between these two covariates. Thus, we will keep both covariates, as removing one of them would worsen the model.

##		GVIF	Df	GVIF ^{1/(2*Df)}
##	<code>total_count</code>	1.013216	1	1.006586
##	<code>age</code>	1.013057	1	1.006507
##	<code>area</code>	2.831994	1	1.682853
##	<code>station</code>	4.594295	1	2.143431
##	<code>as.factor(district)</code>	4.814366	3	1.299441
##	<code>type</code>	3.064633	3	1.205211

We used the function `VIF()`, to detect multicollinearity. We obtained results between the value 1 and 5, which is considered to be generally acceptable. Therefore, multicollinearity is unlikely to be a significant problem.

As we have reached a point where the further reduction of variables might seem illogical, the next step in our view is to consider possible interactions between the covariates to our model. We also decided to concentrate primarily on interactions in which at least one of the variables is categorical, simplifying interpretation.

Study of interactions in the model We propose 5 models with 5 different interactions:

- **age and district:** The influence of a house's age on price may vary by district (older homes might be more desirable in historic areas). [`model_int1`]
- **area and district:** Larger properties could command a higher premium in districts where spacious homes are rare. [`model_int2`]
- **station and district:** Proximity to a train station may be less valuable in certain districts. [`model_int3`]
- **total_count and type:** The number of rooms may affect price differently for various property types (extra rooms in a flat against a detached home). [`model_int4`]
- **area and type:** The effect of square footage on price depends on the property type, reflecting different buyer expectations (for example, space in a flat against a terraced house). [`model_int5`]

These models generally give easily interpretable estimates, indicating statistical significance overall. However, we can note that some of the variables created have relatively high p-values, over 0.05, so not statistically

significant enough. That is why, we will use multiple F-tests, with a standardized 5% significance level, to compare the model_2 and the different models, with the interaction terms mentioned above. The hypotheses are:

Null Hypothesis(H_0): The interaction terms don't improve the model.

Alternative Hypothesis(H_1): At least one of the interaction terms isn't zero, indicating that including these terms in the model improves fit.

We performed this statistical test, using `anova()`. We obtained a variety of outputs, since the p-values are less than 0.05, we can then reject H_0 for multiple models. The ones with the strongest statistical significance are: **age** and **district**, **area** and **district**, and **area** and **type** (from most to less significant). The smallest p-value for model_int1 equal to 1.141e-14, followed by model_int2 (2.546e-13) and then by model_int5 (2.487e-08).

This result indicates that at least one of the coefficient between the interaction isn't equal to zero. Therefore, these models, with the interaction terms, are preferred, as it provides a better fit than model_2.

Previously, we demonstrated that three interactions are statistically significant and sensible. The 3 models seem to be a better choice than the model without interactions (model_2), as we checked through hypothesis testing. They are all easily interpretable and have similar complexity.

However, when looking at results from the hypothesis tests, we see a stronger statistical significance in model_int1 and model_int2 than in model_int5. The adjusted R^2 coefficient for these new models are 0.8984, 0.8971 and 0.892 respectively. These values show once again that model_int5 could be left behind, as the adjusted R^2 coefficient for model_2 is only slightly lower ($0.8839 < 0.892$), indicating minimal return on complexity. The other 2 models show stronger improvement in the R^2 , 0.8984 and 0.8971 against 0.8839, indicating that, despite the added complexity, the new model better accounts for variance in the outcome variable. From both analyses, model_int1 is suggested to be slightly more accurate than model_int2. We conclude that model_int1 is the best option.

Multiple interaction terms in the model To further improve the model, we highlight that the interaction terms present in both model_int1 and model_int2 are not mutually exclusive. It is possible for them to operate simultaneously in a way that strengthen model. We propose constructing a new model that combines the two of them. The model allows us to see whether capturing the two different interaction effects at once yields a more accurate or more interpretable model than using each interaction in isolation.

For constructing this new model, we will incorporate the interaction between **area** and **district** to our model, model_int1:

```
model_doubleint <- lm(price ~ total_count + area * as.factor(district) + type + station
                      + as.factor(district) * age, data = prices)
```

```
##
## Call:
## lm(formula = price ~ total_count + area * as.factor(district) +
##     type + station + as.factor(district) * age, data = prices)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36727  -6342  -1204   4950   70924
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   137574.93    6718.24   20.478  < 2e-16 ***
## total_count    10298.65     277.06   37.171  < 2e-16 ***
```



```
## area                603.59      45.81  13.176 < 2e-16 ***
## as.factor(district)2 52319.47  5875.96   8.904 < 2e-16 ***
## as.factor(district)3 16323.32  7237.65   2.255 0.02456 *
## as.factor(district)4 47575.51  6065.99   7.843 2.84e-14 ***
## typeflat            -15043.53  3237.91  -4.646 4.37e-06 ***
## typesemi_detached   -4007.19  2976.45  -1.346 0.17884
## typeterraced        -9450.89  2713.12  -3.483 0.00054 ***
## station              5191.01   550.73   9.426 < 2e-16 ***
## age                 -1961.57   113.42  -17.295 < 2e-16 ***
## area:as.factor(district)2 -391.43    53.64  -7.298 1.21e-12 ***
## area:as.factor(district)3 -419.15    63.55  -6.596 1.11e-10 ***
## area:as.factor(district)4 -425.25    55.84  -7.616 1.39e-13 ***
## as.factor(district)2:age   390.71   132.56   2.948 0.00336 **
## as.factor(district)3:age   701.31   172.56   4.064 5.63e-05 ***
## as.factor(district)4:age  -431.09   139.49  -3.091 0.00211 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11240 on 483 degrees of freedom
## Multiple R-squared:  0.9144, Adjusted R-squared:  0.9116
## F-statistic: 322.4 on 16 and 483 DF,  p-value: < 2.2e-16
```

The summary of this new model shows a significant increase in the adjusted R^2 , $0.9116 > 0.8984$, since adjusted R-squared penalizes for extra parameters, its improvement here demonstrates that the additional variables contribute meaningfully to the model.

As the previous model is nested in this new one, an F-test is appropriate to properly compare model_int1 and model_doubleint, with a standardized 5% significance level. The hypotheses are:

Null Hypothesis(H_0): The additional interaction terms don't improve the model.

Alternative Hypothesis(H_1): At least one of the additional interaction terms isn't zero, indicating that including these terms in the model significantly improves the fit.

```
## Analysis of Variance Table
##
## Model 1: price ~ total_count + area + station + as.factor(district) *
##      age + type
## Model 2: price ~ total_count + area * as.factor(district) + type + station +
##      as.factor(district) * age
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1     486 7.0557e+10
## 2     483 6.1048e+10  3 9509190719 25.078 4.285e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

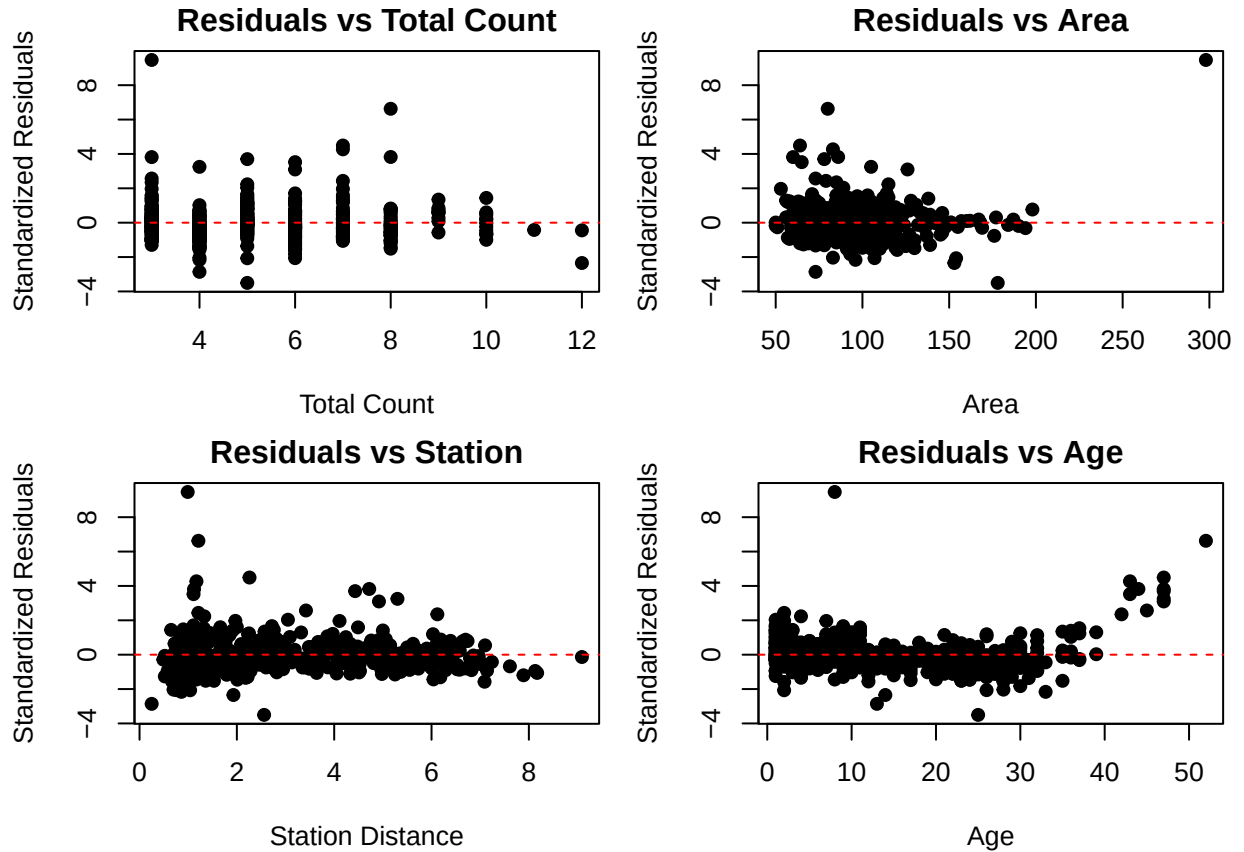
As the p-value of the F-test is very small, $4.285e-15$, it is suggested that the new model (model_intdouble), with multiple interactions, is preferred to the previous model (model_int1), with a single interaction.

Checking for multicollinearity, we once more found no VIF values over 5. Therefore, considering that we manipulated our model, we are confident that multicollinearity isn't a significant issue. Choosing model_doubleint as our final model, we opt to maximize explanatory power. As demonstrated, price of a property depending on age and area vary more conditional on its location (District). By using the variable in both interactions, we add complexity, without excessively compromising the interpretability.

Model checking for final chosen model

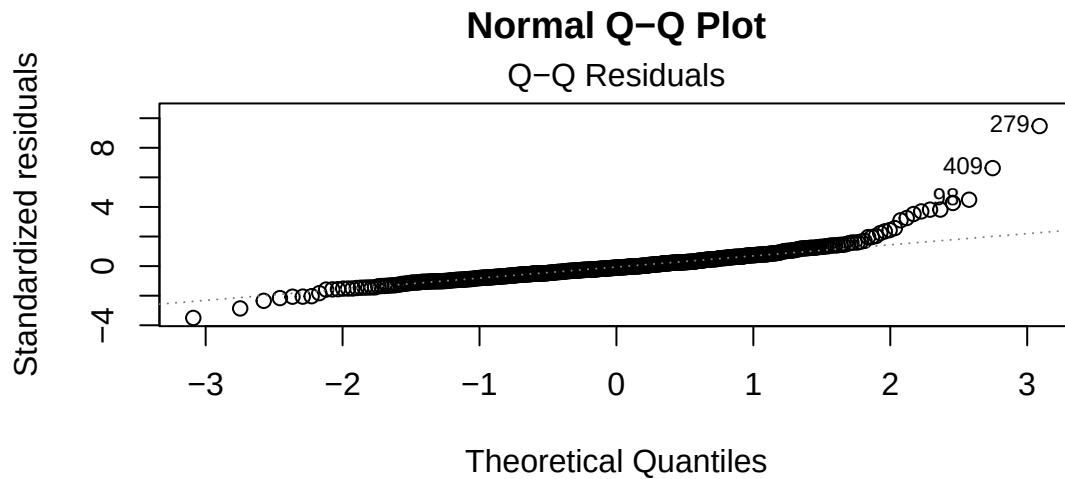
In order to check the final model, we have to evaluate the assumptions of the chosen linear regression model.

Linearity We check for linearity by plotting the standardised residuals against each covariate.



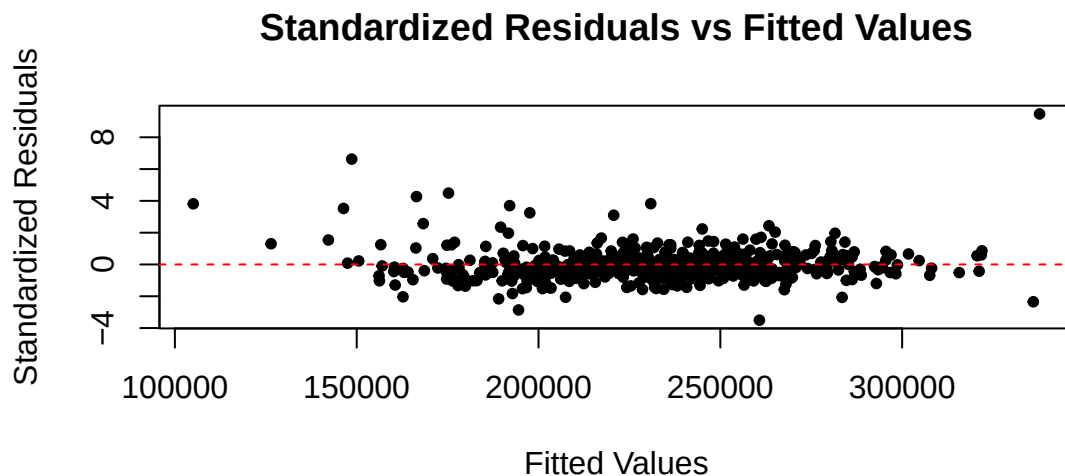
We see that the residuals are scattered randomly around zero for the numerical variables, which suggests that the model can capture linear relationships. Furthermore, we can also see that there is no increasing or decreasing spread in the residuals, indicating constant variance and therefore homoscedasticity.

Normality of residuals We use a Q-Q plot to check the normality of residuals



We can see that the residuals mostly align along the 45 degree reference line, meaning that the residuals are approximately normally distributed. There are some potential outliers at the tails, but they are unlikely to affect the overall model significantly.

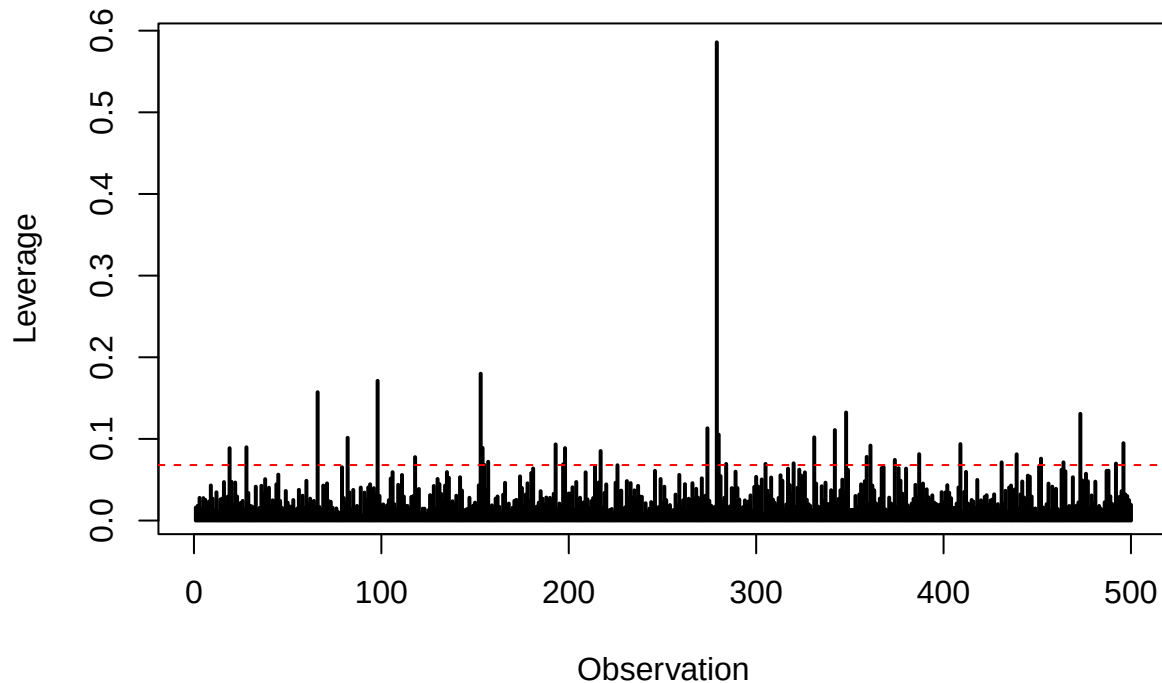
Homoscedasticity We check for a constant variance in the residuals plotting Standardized Residuals against Fitted Values:



We can see on the plot a relatively constant spread of the residuals across all fitted values, which satisfies homoscedasticity.

Independence of error components Checking for the independence of error components, doesn't make sense when there is no information about how the data was collected. The independence of errors is normally evaluated in situations where the data has a natural order. Without knowing whether the data is ordered, testing for autocorrelation is meaningless because the concept of independence depends on assuming that residuals are adjacent in some logical sequence.

Leverage vs. Observation



Lacking information about the data collection, even if from the plot the 279th observation seems to be an outlier and a leverage point, we decide to keep it.

Conclusion

The linear regression model highlights several main factors influencing property sale prices. Location plays a crucial role, as properties in certain districts, such as District 2, exhibit higher prices compared to others. Contrary to what we might expect, the model shows that properties further away from stations attract higher prices. Additionally, older properties prices varies across districts, having generally a negative impact on the property selling price, but this impact being the highest in district 4 and the lowest in district 3. Overall property size, including the total number of rooms and area, positively impacts price, with larger properties commanding higher values. In relation to district, the m^2 seems to be the most expensive in district 1, followed by district 2,3 and 4. Property type also affects value, with flats and terraced houses generally priced lower than detached properties.

Discussion of limitations

A limitation of the model is the lack of knowledge about the data collection. It does not allow us to check for independence, which questions the trustworthiness of the model results and the opportunity to improve it. Furthermore, as we are not able to question the data collection, we are not able to remove points with high leverage, which is a problem as these have a relevant impact on the model. This can lead to misleading outputs, specially as the points don't follow the same pattern as the rest of the data.

Also, as we do not know about the construction history of Statsville, there might be properties built during certain periods of time that are more desirable than others. For example, by looking at the EDA we can see that the properties built 30 years ago, have lower selling prices than the ones built 10 or 45 years ago. So,

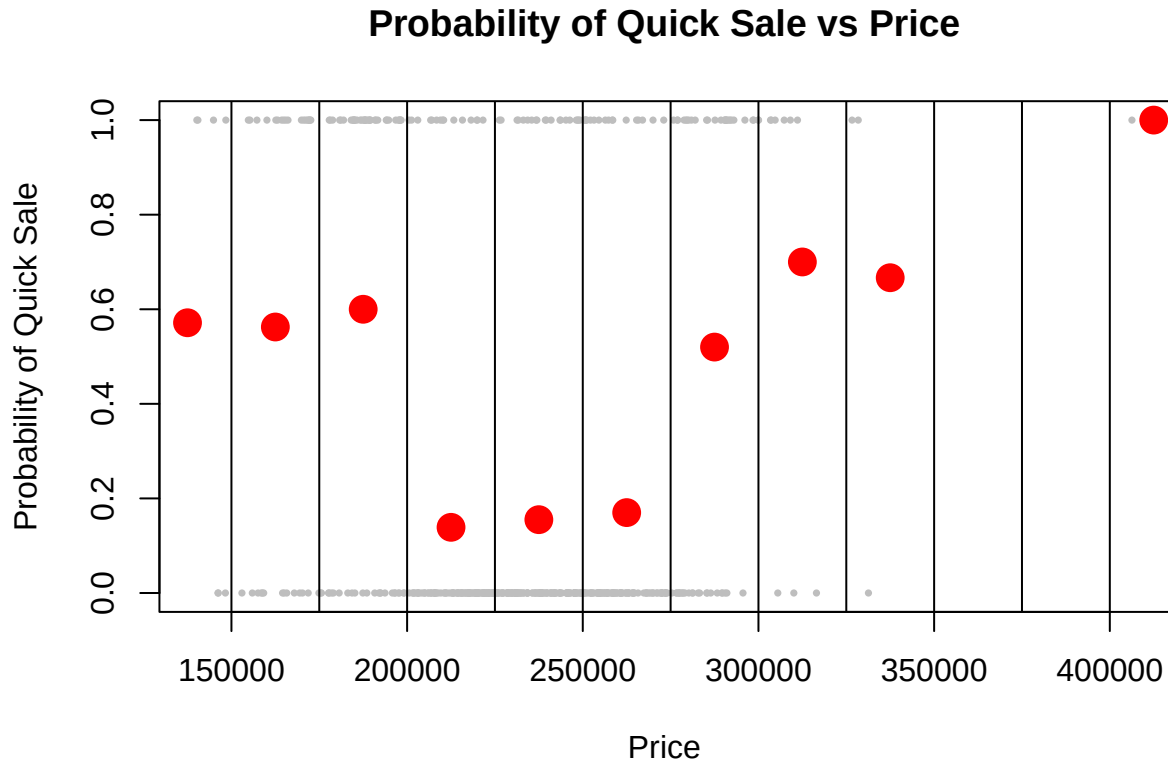
the linear relationship assumption is clearly limited. To solve this issue we could change the covariate from numerical to categorical of several periods of time, ranging from 0 to 5 years, 6 to 10 years... However, the added complexity would probably be too elevated.

Part 2: Generalised linear model OR Generalised additive model

Analysis of sales times

Data analysis

To choose whether to use a GLM or GAM our first thought was to check whether it is reasonable to assume a linear relationship between the covariates and the response. To model the relationship between price and quick_sale we decided to plot the average value for quick_sale for intervals of SD 25000, as shown below. It is not possible to identify any parametric relationship in the plot, therefore we decided to take a non-parametric approach and use a GAM instead of a GLM.



Given that quick_sale is a binary response variable, we will use a Binomial distribution, which belongs to the exponential family, and is a suitable approach for the GAM.

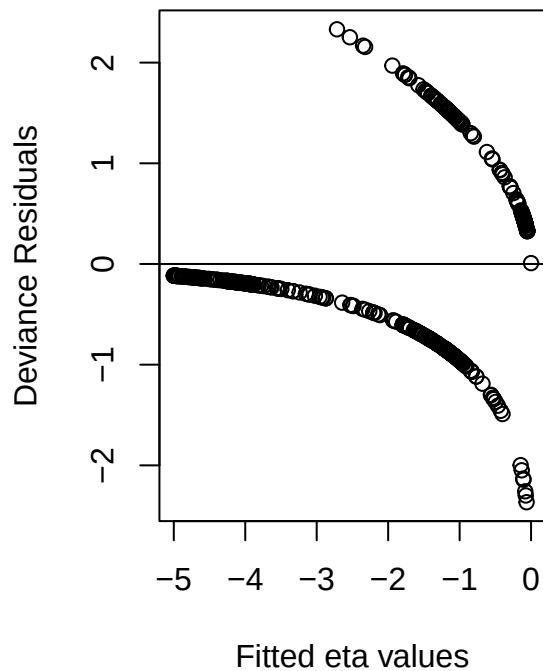
Our objective is to model the probability of a quick sale as a function of the selling price and the agent employed. A critical consideration is ensuring that the probability p_i (which corresponds to μ_i in this context) remains within the interval $[0,1]$. This is typically achieved by applying a logit link function.

$$g(p_i) = \log(p_i / (1 - p_i)) = \eta_i$$

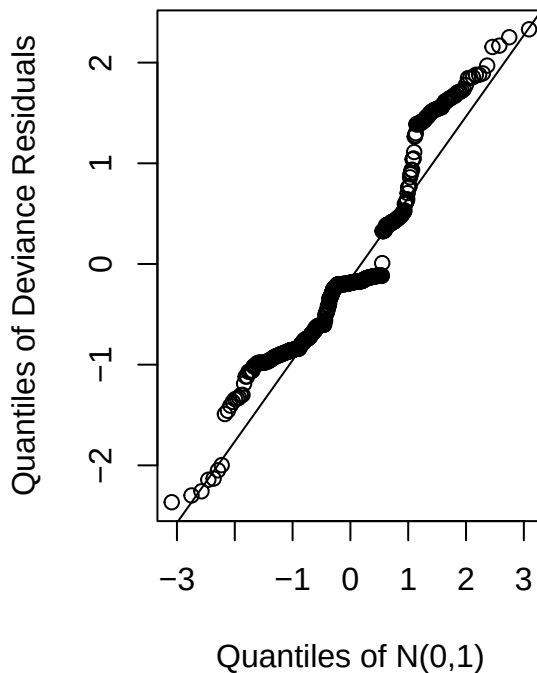
```
times.gam <- gam(quick_sale ~ s(price) + agent, data = times, family = binomial(link="logit"))
```

```
##
## Family: binomial
## Link function: logit
##
## Formula:
## quick_sale ~ agent + s(price)
##
## Parametric coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.2927    0.3274 -10.057  <2e-16 ***
## agentB       3.3818    0.3795   8.911  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##             edf Ref.df Chi.sq p-value
## s(price)  7.974  8.496  78.72  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.422   Deviance explained = 40.7%
## UBRE = -0.24348   Scale est. = 1          n = 500
```

Fitted eta values vs Deviance Residuals

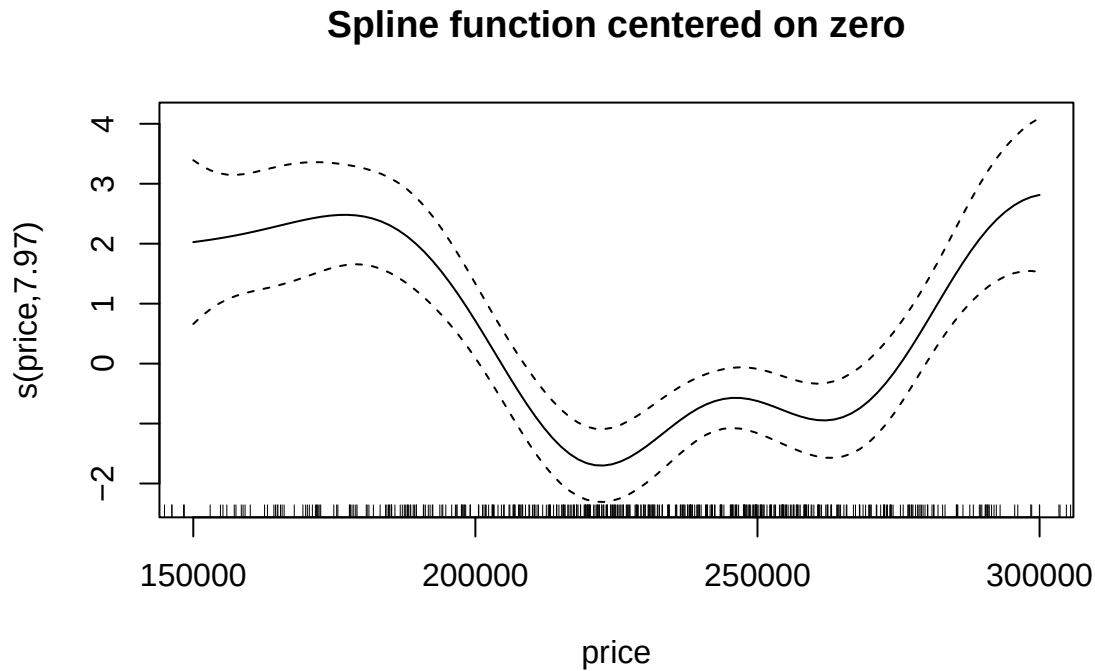


Q-Q plot



The plot checking for constant variance is clearly not one we are familiar with. We found out that this type of result is to be expected for a discrete, binary response variable ¹. There are two lines of points representing the two discrete possible outcomes (0, 1). Given the context, the plot is not worrying. More specifically, as shown from our abline, we have shown that there is a pattern, but there is no systematic non-linear variation of the mean residual. As we don't see a systematic bend of the average residual, there is no violation of our assumption.

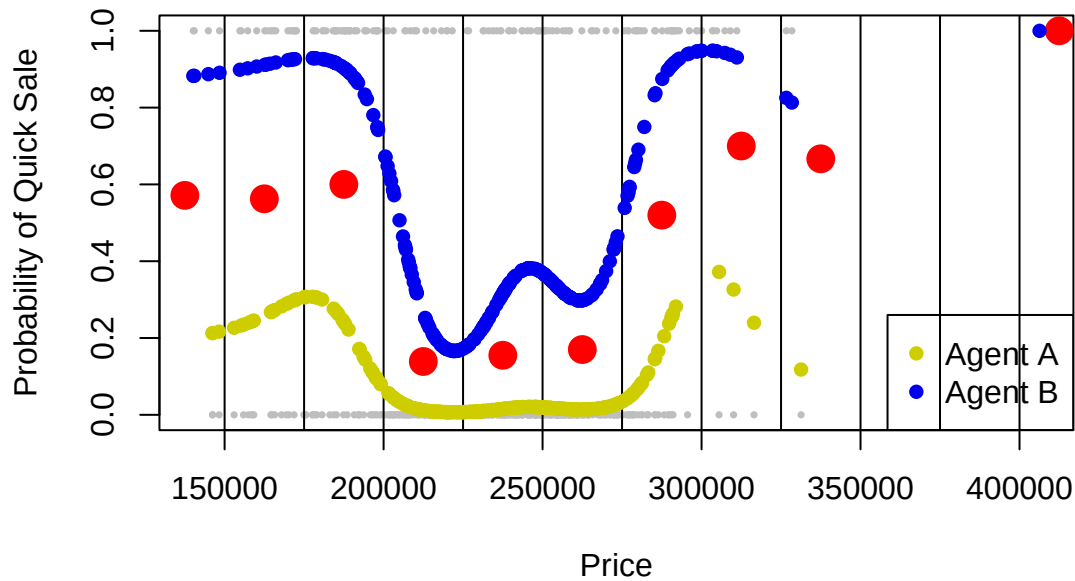
The Q-Q plot shows that the residuals are roughly normally distributed. We are aware that our values are discrete therefore they are not exactly normal, however, we once again can be confident that no assumptions have been violated.



In this plot, we see how the smooth effect of price varies around zero, suggesting that changes in price have an effect, even if it is limited, on the probability of a quick sale. Prices before SD 150,000 and beyond SD 300,000 are not shown as they rise sharply the smooth function and complicates the interpretability of the graph.

¹Gorelik, B. (2012). <https://stats.stackexchange.com/questions/25068/interpreting-plot-of-residuals-vs-fitted-values-from-poisson-regression>

Probability of Quick Sale vs Price



We plotted the fitted values of the model onto the graph of the prices vs the probability of quick sale, to assess how the model captures the relationship between the covariates and the response. The fitted probabilities for both agents seem to follow the trend of the previous empirical results, where agent B has higher overall probabilities.

```
##
## Family: binomial
## Link function: logit
##
## Formula:
## quick_sale ~ s(price) + agent
##
## Parametric coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -3.2927    0.3274 -10.057  <2e-16 ***
## agentB       3.3818    0.3795   8.911  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##             edf Ref.df Chi.sq p-value
## s(price)  7.974  8.496  78.72  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.422   Deviance explained = 40.7%
## UBRE = -0.24348   Scale est. = 1          n = 500
```

The GAM provides valuable insights into the factors influencing the probability of a quick sale. The smooth term for price ($s(\text{price})$) demonstrates a significant non-linear relationship ($p < 2e-16$), indicating a complex

curve. Initially, the adjusted R-squared value of 0.422 raised concerns, as it seemed low compared to typical regression models. However, such values are common for binary response models because some variability remains unexplained. The deviance explained by the model (40.7%) is also consistent with this type of data and model.

The results clearly show that Agent B is far more effective at achieving quick sales compared to Agent A. We find a more complex effect of price, as it doesn't have a simple linear impact but rather influences the probability of a quick sale differently depending on the price range.

```
##           1           2
## 0.0704869 0.6905307
```

Based on the predicted probabilities, the probability of achieving a quick sale for a property at 200,000 SD is approximately 7% with Agent A and 69% with Agent B. Therefore, the manager should choose Agent B to maximize the chances of a quick sale.

Total word count: 2998

Statement about use of generative AI tools

We have not used AI.